



7/21/2026

ASSIGNMENT DAY2

Power Supply Systems and Power Connectors.

TASK 1.

- ❖ **List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.**

➤ The Real World Devices Using SMPS and UPS :

Devices Using SMPS (Switched Mode Power Supply):

- ✓ **Desktop Computer** - It uses SMPS to provide stable power to all the internal parts.
- ✓ **LED TV** - It uses SMPS to convert AC power into DC power efficiently.
- ✓ **Laptop Charger** - It uses SMPS to safely convert high voltage AC into low voltage DC for charging.

Devices Using UPS (Uninterruptible Power Supply):

- ✓ **Device** - Why it uses UPS.
- ✓ **Desktop Computer** - It uses UPS to keep the computer running for a short time during a power cut and protect important data.
- ✓ **WI FI Router** - It uses UPS to keep the internet working during a power outage.

TASK 2.

- ❖ Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections. Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io.

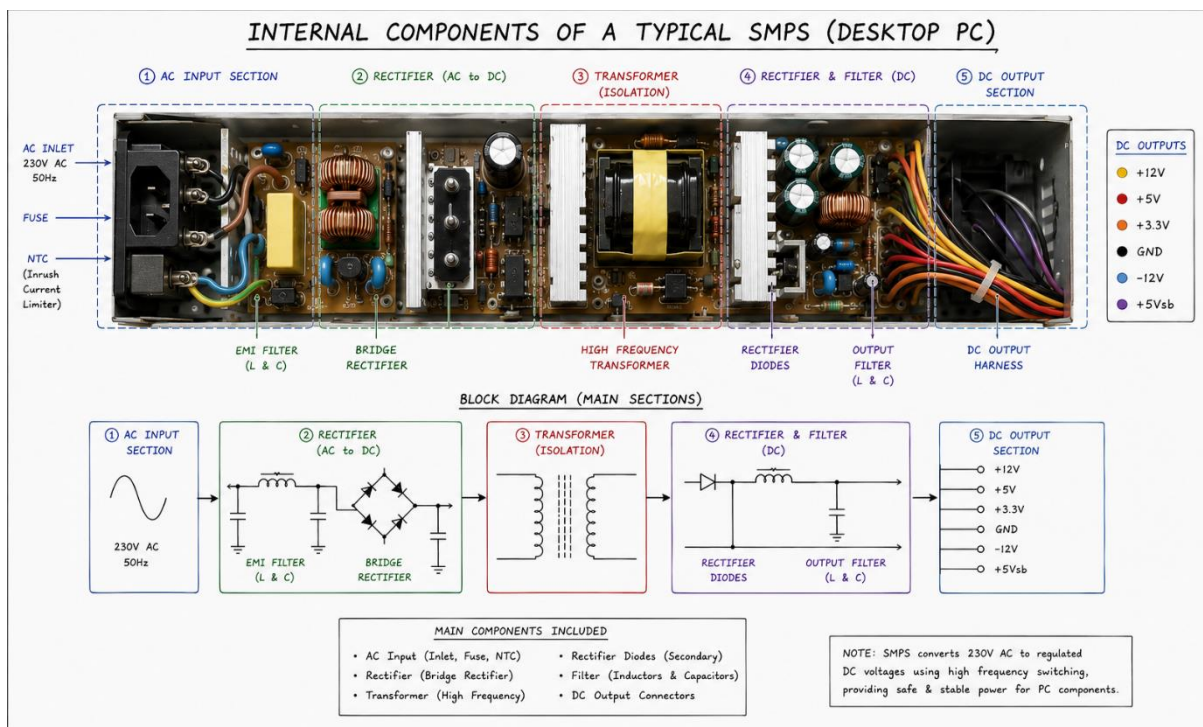


Figure : Internal Components of a Typical Desktop PC SMPS (ATX).

Explanation :

- ✓ The above diagram shows the main internal components of a typical SMPS (Switched Mode Power Supply) used in a desktop PC. It receives AC power from the mains supply and converts it into DC power. The rectifier changes AC into DC, the high-frequency transformer adjusts the

voltage safely, and the DC output section provides stable voltages such as +3.3V, +5V, and +12V to the motherboard, processor, storage devices, and other computer components. This ensures that the PC receives a safe and reliable power supply.

TASK 3.

- ❖ Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

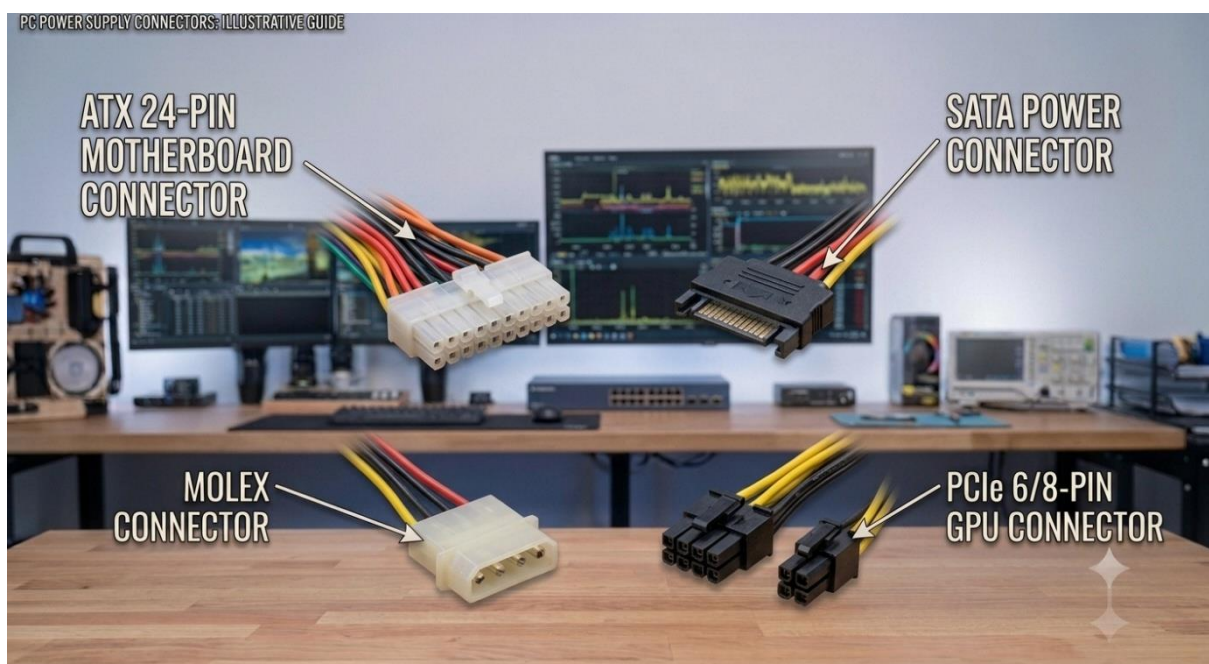


Figure : Common PC Power Supply Connectors (ATX 24 Pin, SATA Power, Molex and PCIe 6/8 Pin).

Explanation :

- ✓ The image shows four common power supply connectors used inside a desktop computer. Each connector has a different purpose and provides

power to specific hardware components. Using the correct connector and handling it carefully helps the computer work safely and efficiently.

Connector Details

1. ATX 24 Pin Motherboard Connector:

- ✓ Devices it powers: Motherboard (main power supply)
- ✓ Safety Tip: Turn off the power supply before connecting or disconnecting it. Make sure the connector is fully locked.

2. SATA Power Connector:

- ✓ Devices it powers: SATA SSDs, HDDs, and optical drives
- ✓ Safety Tip: Insert and remove the connector gently. Never force it into the port.

3. Molex Connector:

- ✓ Devices it powers: Older hard drives, cooling fans, optical drives, and other peripherals
- ✓ Safety Tip: Hold the connector while removing it. Do not pull the wires.

TASK 4.

- ❖ **Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.**

➤ **Connecting the SMPS to a New Gaming PC:**

Step 1: Turn off the Power Supply

- ✓ First, switch OFF the SMPS and unplug the power cable. This keeps the installation safe.

Step 2: Connect the Motherboard

- ✓ Connect the 24-pin ATX power cable from the SMPS to the 24-pin motherboard connector. This provides the main power to the motherboard.

Step 3: Connect the Graphics Card (GPU)

- ✓ Connect the PCIe 6/8-pin power cable from the SMPS to the PCIe power connector on the graphics card. This supplies the extra power needed for gaming.

Step 4: Connect the Storage Devices

- ✓ Connect the SATA power cable from the SMPS to the SATA power connector on the SSD or HDD. This supplies power to the storage devices.

Step 5: Check All Connections

- ✓ Make sure all cables are connected properly and locked in place. Do not force any connector.

Step 6: Turn On the PC

- ✓ Plug in the power cable, switch ON the SMPS, and start the computer. Check that the motherboard, graphics card, and storage devices are working properly.